

Statement of Work

Project title : Online Sales Dashboard

Objective

The objective of this project is to clean, analyze, and visualize online retail transaction data in order to understand sales performance, customer behavior, product performance, country-level revenue, and cancellation/return patterns.

The final deliverable will be an interactive Excel sales dashboard that helps business stakeholders answer key questions about revenue, orders, products, customers, and markets.

Business Problem

The business needs a clear view of online retail sales performance. The raw transaction data contains invoice-level product purchases, but it is difficult to understand performance trends without cleaning, organizing, and summarizing the data.

This project will turn raw transaction data into a dashboard that can help answer:

What is selling well, where revenue is coming from, who the best customers are, and where the business may be losing money through cancellations or returns?

Dataset Description

The dataset contains online retail transaction records. Each row represents one product line inside an invoice.

The main columns are:

Column	Description
Invoice	Unique invoice/order number
StockCode	Product code
Description	Product name
Quantity	Number of units purchased
InvoiceDate	Date and time of transaction

Price	Unit price
Customer ID	Unique customer identifier
Country	Customer country

Important note: One invoice can appear multiple times because one customer order may contain many different products.

Project Scope

This project will include the following phases:

Phase	Description
Data Collection	Import and review the raw online retail dataset
Data Cleaning	Fix data types, handle missing values, identify cancellations, returns, invalid prices, and duplicates
Data Analysis	Create calculated fields, pivot tables, and summary metrics
Data Visualization	Build an interactive Excel dashboard with KPIs, charts, and slicers
Reporting	Document the process, methods, findings, and dashboard features
Business Insights	Provide clear insights and recommendations based on the analysis

Out of Scope

To keep the project focused, this first version will not include:

Out of Scope Item	Reason
Predictive modeling	This project focuses on Excel dashboarding and descriptive analytics
Machine learning	Not needed for the first portfolio version
Real-time dashboard updates	The dataset is static

Advanced customer segmentation	Can be added later as a second project phase
Profit analysis	Dataset has revenue but does not include cost or profit data

Tools Used

Tool	Purpose
Microsoft Excel	Main tool for cleaning, analysis, and dashboarding
Excel Tables	Organizing the dataset
Formulas	Creating calculated fields
Pivot Tables	Summarizing sales, customers, products, and countries
Pivot Charts	Creating visual reports
Slicers	Making the dashboard interactive
Power Query	For stronger cleaning workflow

Key Business Questions

The dashboard will answer these questions:

Question	Dashboard Output
What is total revenue?	Total Revenue KPI
How many orders were placed?	Total Orders KPI
How many customers purchased?	Total Customers KPI
What is the average order value?	AOV KPI
Which months generated the most revenue?	Monthly revenue trend

Which countries generated the most revenue?	Revenue by country chart
Which products generated the most revenue?	Top products chart
Who are the highest-value customers?	Top customers table
How many orders were cancelled?	Cancelled orders KPI/chart
Which days of the week perform best?	Revenue by weekday chart

Deliverables

At the end of the project, we will have:

Deliverable	Description
Cleaned Dataset	A cleaned version of the original transaction data
Calculated Columns	Revenue, order status, transaction type, month, year, weekday, hour
Pivot Tables	Summary tables for revenue, products, customers, countries, and cancellations
Excel Dashboard	Interactive dashboard with KPIs, charts, and slicers
Project Documentation	Explanation of cleaning steps, analysis process, and insights

Success Criteria

This project will be considered successful if:

Criteria	Description
Clean data	Dataset is formatted correctly and major data issues are addressed

Clear analysis	Pivot tables answer the main business questions
Professional dashboard	Dashboard is clean, readable, and interactive
Useful insights	Findings are explained in simple business language